MSG No. 213.64 $P4_1321'$ [Type II, cubic]

Table 1: Wyckoff site: 4a, site symmetry: .321'

No.	position	mapping
1	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	[1,12,14,16,17,18,25,36,38,40,41,42]
2	$\left[\frac{5}{8}, \frac{7}{8}, \frac{1}{8}\right]$	[2,7,9,15,19,24,26,31,33,39,43,48]
3	$\left[\frac{1}{8}, \frac{5}{8}, \frac{7}{8}\right]$	[3,4,10,11,20,21,27,28,34,35,44,45]
4	$\left[\frac{7}{8}, \frac{1}{8}, \frac{5}{8}\right]$	[5,6,8,13,22,23,29,30,32,37,46,47]

Table 2: Wyckoff site: 4b, site symmetry: .321'

No.	position	mapping
1	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	[1,12,14,16,17,18,25,36,38,40,41,42]
2	$\left[\frac{1}{8}, \frac{3}{8}, \frac{5}{8}\right]$	[2,7,9,15,19,24,26,31,33,39,43,48]
3	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{8}\right]$	[3,4,10,11,20,21,27,28,34,35,44,45]
4	$\left[\frac{3}{8}, \frac{5}{8}, \frac{1}{8}\right]$	[5,6,8,13,22,23,29,30,32,37,46,47]

Table 3: Wyckoff site: 8c, site symmetry: .3.1'

No.	position	mapping
1	$[x, x, x]$	[1,17,18,25,41,42]
2	$\left[x+\frac{1}{4}, \frac{1}{4}-x, x+\frac{3}{4}\right]$	[2,7,15,26,31,39]
3	$\left[x+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{4}-x\right]$	[3,4,11,27,28,35]
4	$\left[\frac{1}{4}-x, x+\frac{3}{4}, x+\frac{1}{4}\right]$	[5,6,13,29,30,37]
5	$\left[x+\frac{1}{2}, \frac{1}{2}-x, -x\right]$	[8,22,23,32,46,47]
6	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	[9,19,24,33,43,48]
7	$\left[\frac{1}{2}-x, -x, x+\frac{1}{2}\right]$	[10,20,21,34,44,45]
8	$\left[\frac{3}{4}-x, \frac{3}{4}-x, \frac{3}{4}-x\right]$	[12,14,16,36,38,40]

Table 4: Wyckoff site: 12d, site symmetry: ..21'

No.	position	mapping
1	$\left[\frac{1}{8}, y, y+\frac{1}{4}\right]$	[1,13,25,37]
2	$\left[\frac{3}{8}, -y, y+\frac{3}{4}\right]$	[2,10,26,34]
3	$\left[\frac{7}{8}, y+\frac{1}{2}, \frac{1}{4}-y\right]$	[3,9,27,33]
4	$\left[y, y+\frac{1}{4}, \frac{1}{8}\right]$	[4,18,28,42]
5	$\left[-y, y+\frac{3}{4}, \frac{3}{8}\right]$	[5,19,29,43]
6	$\left[\frac{1}{4}-y, \frac{7}{8}, y+\frac{1}{2}\right]$	[6,20,30,44]
7	$\left[y+\frac{1}{4}, \frac{1}{8}, y\right]$	[7,17,31,41]
8	$\left[\frac{5}{8}, \frac{1}{2}-y, \frac{3}{4}-y\right]$	[8,14,32,38]
9	$\left[y+\frac{3}{4}, \frac{3}{8}, -y\right]$	[11,22,35,46]
10	$\left[\frac{3}{4}-y, \frac{5}{8}, \frac{1}{2}-y\right]$	[12,24,36,48]
11	$\left[y+\frac{1}{2}, \frac{1}{4}-y, \frac{7}{8}\right]$	[15,23,39,47]
12	$\left[\frac{1}{2}-y, \frac{3}{4}-y, \frac{5}{8}\right]$	[16,21,40,45]

Table 5: Wyckoff site: `24e`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,25]`
2	$\left[x+\frac{1}{4},\ \frac{1}{4}-z,\ y+\frac{3}{4}\right]$	`[2,26]`
3	$\left[x+\frac{3}{4},\ z+\frac{1}{4},\ \frac{1}{4}-y\right]$	`[3,27]`
4	$\left[z+\frac{3}{4},\ y+\frac{1}{4},\ \frac{1}{4}-x\right]$	`[4,28]`
5	$\left[\frac{1}{4}-z,\ y+\frac{3}{4},\ x+\frac{1}{4}\right]$	`[5,29]`
6	$\left[\frac{1}{4}-y,\ x+\frac{3}{4},\ z+\frac{1}{4}\right]$	`[6,30]`
7	$\left[y+\frac{1}{4},\ \frac{1}{4}-x,\ z+\frac{3}{4}\right]$	`[7,31]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[8,32]`
9	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[9,33]`
10	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	`[10,34]`
11	$\left[y+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{4}-z\right]$	`[11,35]`
12	$\left[\frac{3}{4}-y,\ \frac{3}{4}-x,\ \frac{3}{4}-z\right]$	`[12,36]`
13	$\left[\frac{1}{4}-x,\ z+\frac{3}{4},\ y+\frac{1}{4}\right]$	`[13,37]`
14	$\left[\frac{3}{4}-x,\ \frac{3}{4}-z,\ \frac{3}{4}-y\right]$	`[14,38]`
15	$\left[z+\frac{1}{4},\ \frac{1}{4}-y,\ x+\frac{3}{4}\right]$	`[15,39]`
16	$\left[\frac{3}{4}-z,\ \frac{3}{4}-y,\ \frac{3}{4}-x\right]$	`[16,40]`
17	$[z,\ x,\ y]$	`[17,41]`
18	$[y,\ z,\ x]$	`[18,42]`
19	$\left[-y,\ z+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[19,43]`
20	$\left[\frac{1}{2}-z,\ -x,\ y+\frac{1}{2}\right]$	`[20,44]`
21	$\left[\frac{1}{2}-y,\ -z,\ x+\frac{1}{2}\right]$	`[21,45]`
22	$\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ -y\right]$	`[22,46]`
23	$\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ -x\right]$	`[23,47]`
24	$\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[24,48]`